

The Chi-squared Test

Rex Shao-Yu Wang

Department of Agronomy, National Taiwan University

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Outline

Quick Review

Building the Data

R Implementation

Paired Data: McNemar Test

Summary

When Do We Use the Chi-squared Test?

The Decision Rule

Both the response (Y) **and** the predictor (X) are **categorical** (counts / frequencies, not means).

Data type	Test
Continuous Y, continuous X	Correlation / Regression
Continuous Y, categorical X	t-test / ANOVA
Categorical Y, categorical X	Chi-squared test

Core Idea

The core idea is always: compare **Observed** vs **Expected** counts.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Three Chi-squared Tests — One Function in R

- ▶ **Test of Independence**

Tests whether two categorical variables are associated

- ▶ **Goodness-of-Fit Test**

Tests whether observed frequencies match a specified distribution

- ▶ **Test of Homogeneity**

Tests whether multiple populations share the same distribution

Test	Question	R function
Goodness-of-Fit	Fit a theoretical ratio?	<code>chisq.test(x, p=)</code>
Independence	Two variables associated?	<code>chisq.test(table)</code>
Homogeneity	Same distribution across groups?	<code>chisq.test(table)</code>

Two Ways to Build a Contingency Table

Option A — Type the counts directly with `matrix()`

```
oil <- matrix(c(154, 132,  
               180, 126,  
               104, 131),  
             nrow = 3, byrow = TRUE)  
rownames(oil) <- c("Support", "Oppose", "DoNotKnow")  
colnames(oil) <- c("Grad", "NonGrad")
```

Option B — From raw data with `table()`

```
# df has one row per respondent  
oil <- table(df$opinion, df$college)
```

Goodness-of-Fit Test

What it tests

Whether the observed counts of a **single categorical variable** match a **theoretical / expected ratio**.

$$\begin{cases} H_0 : \text{the population follows the given distribution} \\ H_1 : \text{it does not} \end{cases}$$

- ▶ Expected count: $E_i = p_i \times N$ (a known ratio, no second variable)
- ▶ $df = k - 1$, where k = number of categories

Keep in mind

Here the expected proportions come from **theory** (e.g. a genetic law), not from the data itself.

Goodness-of-Fit Test

Example: Do budgie colours follow Mendel's **9:3:3:1** ratio?

Counts of $N = 971$ birds: Green 550, Blue 168, Yellow 180, White 73.
($\alpha = 0.05$)

```
observed <- c(Green=550, Blue=168, Yellow=180, White=73)
expected_p <- c(9, 3, 3, 1) / 16

chisq.test(x = observed, p = expected_p)
```

X-squared = 3.6342, df = 3, p-value = 0.3038

$\therefore p > 0.05 \therefore$ data are consistent with Mendel's laws.

Test of Independence

What it tests

Whether **two categorical variables** measured on **one sample** are associated (related) or independent.

$$\begin{cases} H_0 : \text{the two factors are independent} \\ H_1 : \text{the two factors are associated} \end{cases}$$

- ▶ Data are arranged in a **contingency table**
- ▶ Expected count: $E_{ij} = (R_i \times C_j)/N$ — estimated **from the data**
- ▶ $df = (r - 1)(c - 1)$

Keep in mind

Unlike goodness-of-fit, there is **no theoretical distribution** — expected counts come from the row and column totals.

Test of Independence

Example: 827 people were asked their opinion on oil drilling, split by college status. Is opinion **associated** with having a degree?

```
oil <- matrix(c(154, 132,
               180, 126,
               104, 131),
             nrow = 3, byrow = TRUE)

chisq.test(oil, correct = FALSE)
chisq.test(oil)$expected
```

X-squared = 11.47, df = 2, p-value = 0.0032

$\therefore p < 0.05 \therefore$ reject H_0 : the two factors are NOT independent.

Reminder

For a **2x2** table, `chisq.test()` applies Yates' correction by **default**. Use `correct = FALSE` to turn it off.

Test of Homogeneity

What it tests

Whether **two (or more) separate populations** share the **same distribution** over the categories of one variable.

$$\begin{cases} H_0 : \text{the populations follow the same distribution} \\ H_1 : \text{at least one population differs} \end{cases}$$

- ▶ Same contingency table, same $E_{ij} = (R_i \times C_j)/N$, same df

Independence vs Homogeneity

The **computation is identical** to the independence test.

The difference is **study design**:

- ▶ Independence: **one** sample cross-classified
- ▶ Homogeneity: **separate** samples from each population

Test of Homogeneity

Example: Separate samples of male and female students were asked their living arrangement. Do the two groups share the **same distribution**?

```
living <- matrix(c(72, 84, 49, 45,
                  91, 86, 88, 35),
                nrow = 2, byrow = TRUE)
rownames(living) <- c("Male", "Female")
colnames(living) <- c("Dorm", "Apt", "Home", "Other")

chisq.test(living)
```

X-squared = 10.13, df = 3, p-value = 0.0175

$\therefore p < 0.05 \therefore$ reject H_0 : the distributions are NOT the same.

Interpreting the R Output

A typical `chisq.test()` output gives three numbers:

- ▶ **X-squared** — the test statistic χ_0^2
- ▶ **df** — degrees of freedom
- ▶ **p-value** — probability of this result if H_0 were true

Decision Rule

If **p-value** $< \alpha$ (significance level) $\Rightarrow$ **reject** H_0 .

McNemar Test

When to use

When the two samples are **paired / not independent** — the **same subjects** measured twice (e.g. the same voters before vs after a debate).

- ▶ H_0 : the proportion is unchanged (no shift between the two times)
- ▶ Only the **off-diagonal** cells (n_{12} , n_{21}) carry information — the agreements on the diagonal cancel out
- ▶ Test statistic: $\chi^2 = \frac{(n_{12} - n_{21})^2}{n_{12} + n_{21}}$, $df = 1$

Why not the ordinary test?

The ordinary chi-squared test assumes **independent** observations. Paired data violate this, so a different statistic is needed.

McNemar Test

Example: 1000 voters stated their choice **before** and **after** a debate.
Did the debate change their decision?

	After: A	After: B
Before: A	491	9
Before: B	1	499

```
votes <- matrix(c(491, 9,  
                  1, 499),  
               nrow = 2, byrow = TRUE)  
  
mcnemar.test(votes, correct = FALSE)
```

McNemar's chi-squared = 6.4, df = 1, p-value = 0.0114

$\therefore p < 0.05 \therefore$ reject H_0 : the debate did change voters' decisions.

Summary: The R Workflow

1. Build the table — `matrix()` or `table()`
2. Run the test — `chisq.test()`
3. **Check** expected counts — `result$expected`
4. Read the p-value and decide

Situation	Function
Fit a ratio	<code>chisq.test(x, p=)</code>
Independence / Homogeneity	<code>chisq.test(table)</code>
Paired data	<code>mcnemar.test()</code>

Same idea every time: Observed vs Expected.

Thank you for your attention!